

(a) Charge Q above grounded plane (b) Equivalent configuration

Figure 4-30 By image theory, a charge Q above a grounded, perfectly conducting plane is equivalent to Q and its image $-Q$ with the grounded plane removed.

► Any given charge configuration above an infinite, perfectly conducting plane is electrically equivalent to the combination of the given charge configuration and its image configuration with the conducting plane removed. ◄

The image-method equivalent of the charge Q above a conducting plane is shown in the right-hand section of **Fig. 4-30**. It consists of the charge Q itself and an image charge $-Q$ at a distance $2d$ from Q with nothing else between them. The electric field due to the two isolated charges can now be easily found at any point (x,y,z) by applying Coulomb's method, as demonstrated by Example 4-17. By symmetry, the combination of the two charges always produces a potential $V = 0$ at every point in the plane previously occupied by the conducting surface. If the charge resides in the presence of more than one grounded plane, it is necessary to establish its images relative to each of the planes and then to establish images of each of those images against the remaining planes.

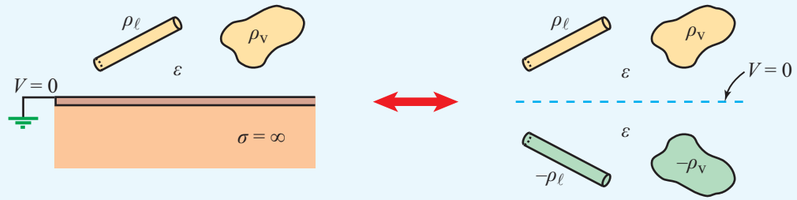
The process is continued until the condition $V = 0$ is satisfied everywhere on all grounded planes. The image method applies not only to point charges but also to distributions of charge, such as the line and volume distributions depicted in **Fig. 4-31**. Once $\mathbf{E}$ has been determined, the charge induced on the plate can be found from

$$\rho_s = (\hat{\mathbf{n}} \cdot \mathbf{E})\varepsilon_0, \tag{4.145}$$

where $\hat{\mathbf{n}}$ is the normal unit vector to the plate (**Fig. 4-30(a)**).

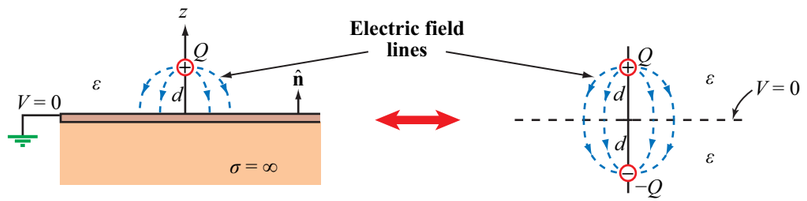
Example 4-17: Image Method for Charge above Conducting Plane

Use image theory to determine $\mathbf{E}$ at an arbitrary point $P = (x,y,z)$ in the region $z > 0$ due to a charge Q in free space at a distance d above a grounded conducting plate residing in the $z = 0$ plane.



(a) Charge distributions above grounded plane (b) Equivalent distributions

Figure 4-31 Charge distributions above a conducting plane and their image-method equivalents.



(a) Charge Q above grounded plane (b) Equivalent configuration

图 4-30 根据镜像理论，接地、理想导电平面上方的电荷 Q 等价于 Q 及其镜像 –Q（除去接地平面）。

► 无限、理想导电平面上方的任何给定电荷配置在电学上等效于给定电荷配置及其去除导电平面后的镜像配置的组合。◄

导电平面上方电荷 Q 的镜像法等效如图 4-30 的右侧部分所示。它由电荷 Q 本身和距 Q 2d 的镜像电荷 −Q 组成，它们之间没有其他任何东西。现在，通过应用库仑法，可以轻松地在任意点 (x, y, z) 找到由两个隔离电荷产生的电场，如例 4-17 所示。根据对称性，两个电荷的组合总是在先前由导电表面占据的平面中的每个点产生电势 V = 0 。如果电荷存在于多个接地平面中，则有必要建立其相对于每个平面的镜像，然后建立每个镜像相对于其余平面的镜像。

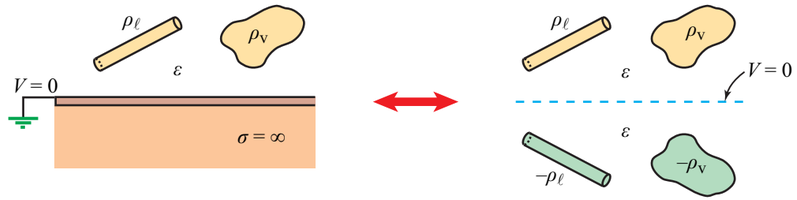
继续该过程，直到所有接地平面上的所有位置都满足条件 V = 0。镜像法不仅适用于点电荷，也适用于电荷分布，如图4-31所示的线分布和体积分布。一旦 E 确定，板上感应的电荷可以由下式找到：

$$\rho_s = (\hat{\mathbf{n}} \cdot \mathbf{E})\varepsilon_0, \tag{4.145}$$

其中n̂是板的法向单位矢量（图4-30(a)）。

例 4-17：导电平面上方电荷的镜像法

使用镜像理论来确定 z > 0 区域中任意点 P = (x, y, z) 处的 E，这是由于位于 z = 0 平面中的接地导电板上方距离 d 处的自由空间中的电荷 Q 造成的。



(a) Charge distributions above grounded plane (b) Equivalent distributions

图 4-31 导电平面上方的电荷分布及其等效镜像方法。

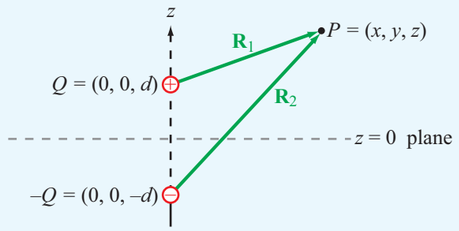


Figure 4-32 Application of the image method for finding **E** at point P (Example 4-17).

Solution: In **Fig. 4-32**, charge Q is at $(0,0,d)$, and its image $-Q$ is at $(0,0,-d)$. From Eq. (4.19), the electric field at point $P(x,y,z)$ due to the two charges is given by the following equation.

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q\mathbf{R}_1}{R_1^3} + \frac{-Q\mathbf{R}_2}{R_2^3}\right)$$
$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{\hat{\mathbf{x}}x+\hat{\mathbf{y}}y+\hat{\mathbf{z}}(z-d)}{[x^2+y^2+(z-d)^2]^{3/2}} - \frac{\hat{\mathbf{x}}x+\hat{\mathbf{y}}y+\hat{\mathbf{z}}(z+d)}{[x^2+y^2+(z+d)^2]^{3/2}}\right]$$

for $z \geq 0$.

Concept Question 4-28: What is the fundamental premise of the image method?

Concept Question 4-29: Given a charge distribution, what are the various approaches described in this chapter for computing the electric field **E** at a given point in space?

Exercise 4-19: Use the result of Example 4-17 to find the surface charge density ρ_s on the surface of the conducting plane.

Answer: $\rho_s = -Qd/[2\pi(x^2+y^2+d^2)^{3/2}]$. (See EM.)

Chapter 4 Summary

Concepts

- Maxwell's equations are the fundamental tenets of electromagnetic theory.
- Under static conditions, Maxwell's equations separate into two uncoupled pairs with one pair pertaining to electrostatics and the other to magnetostatics.
- Coulomb's law provides an explicit expression for the electric field due to a specified charge distribution.
- Gauss's law states that the total electric field flux through a closed surface is equal to the net charge enclosed by the surface.
- The electrostatic field **E** at a point is related to the electric potential V at that point by $\mathbf{E} = -\nabla V$ with V often being referenced to zero at infinity.
- Because most metals have conductivities on the order of 10^6 (S/m), they are treated in practice as perfect conductors. By the same token, insulators with conductivities smaller than 10^{-10} (S/m) often are treated as perfect dielectrics.
- Boundary conditions at the interface between two materials specify the relations between the normal and tangential components of **D**, **E**, and **J** in one of the materials to the corresponding components in the other.
- The capacitance of a two-conductor body and resistance of the medium between them can be computed from knowledge of the electric field in that medium.
- The electrostatic energy density stored in a dielectric medium is $w_e = \frac{1}{2}\varepsilon E^2$ (J/m^3).
- When a charge configuration exists above an infinite, perfectly conducting plane, the induced field **E** is the same as that due to the configuration itself and its image with the conducting plane removed.

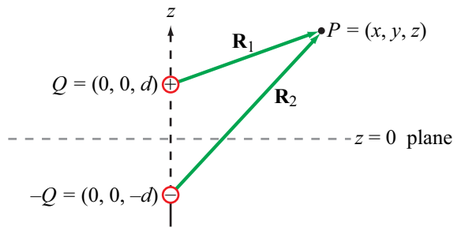


图 4-32 应用镜像法求 P 点 E（例 4-17）。

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q\mathbf{R}_1}{R_1^3} + \frac{-Q\mathbf{R}_2}{R_2^3}\right)$$
$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{\hat{\mathbf{x}}x+\hat{\mathbf{y}}y+\hat{\mathbf{z}}(z-d)}{[x^2+y^2+(z-d)^2]^{3/2}} - \frac{\hat{\mathbf{x}}x+\hat{\mathbf{y}}y+\hat{\mathbf{z}}(z+d)}{[x^2+y^2+(z+d)^2]^{3/2}}\right]$$

解：在图4-32中，电荷Q在(0,0,d)处，其像−Q在(0,0,−d)处。从方程。根据 (4.19) 式，两个电荷在点 P(x, y, z) 处的电场由以下方程给出。

对于 $z \geq 0$。

概念题4-28：镜像法的基本前提是什么？

概念问题 4-29：给定电荷分布，本章描述的计算空间给定点电场 E 的各种方法有哪些？

练习4-19：利用例4-17的结果求导电平面表面的表面电荷密度ρ_s。

答案：$\rho_s = -Qd/[2\pi(x^2+y^2+d^2)^{3/2}]$。（参见EM。）

第四章总结

概念

- 麦克斯韦方程组是电磁理论的基本原理。
- 在静态条件下，麦克斯韦方程分为两对非耦合方程，一对涉及静电学，另一对涉及静磁学。
- 库仑定律为特定电荷分布产生的电场提供了明确的表达式。
- 高斯定律指出，通过封闭表面的总电通量 等于该表面所包围的净电荷。
- 某一点的静电场 E 与该点的电势 V 的关系为 E $= -\nabla V$，其中 V 通常在无穷远处为零。
- 多数金属的电导率约为10^6 S/m，在本章近似下通常视为理想导体。同理，电导率低于 10^{-10} S/m 的绝缘体通常近似视为理想电介质。
- 两种材料之间界面处的边界条件指定了一种材料中 D、E 和 J 的法向和切向分量与另一种材料中相应分量之间的关系。
- 双导体体的电容和它们之间的介质的电阻可以根据该介质中的电场的知识来计算。
- 存储在介电介质中的静电能量密度为$w_e = \frac{1}{2}\varepsilon E^2$ (J/m^3)。
- 当电荷构型存在于无限的、理想的导电平面上方时，感应场 E 与构型本身及其镜像（除去导电平面后）产生的场相同。

Mathematical and Physical Models

Maxwell's Equations for Electrostatics

Name	Differential Form	Integral Form
Gauss's law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
Kirchhoff's law	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

Electric Field

Current density	$\mathbf{J} = \rho_v \mathbf{u}$
Poisson's equation	$\nabla^2 V = -\dfrac{\rho_v}{\varepsilon}$
Laplace's equation	$\nabla^2 V = 0$
Resistance	$R = \dfrac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$
Boundary conditions	Table 4-3
Capacitance	$C = \dfrac{\int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}}$
RC relation	$RC = \dfrac{\varepsilon}{\sigma}$
Energy density	$w_e = \frac{1}{2}\varepsilon E^2$

Point charge	$\mathbf{E} = \hat{\mathbf{R}} \dfrac{q}{4\pi\varepsilon R^2}$
Many point charges	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \sum_{i=1}^{N} \dfrac{q_i(\mathbf{R} - \mathbf{R}_i)}{\lvert\mathbf{R} - \mathbf{R}_i\rvert^3}$
Volume distribution	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{v'} \hat{\mathbf{R}}' \dfrac{\rho_v \, dv'}{R'^2}$
Surface distribution	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{S'} \hat{\mathbf{R}}' \dfrac{\rho_s \, ds'}{R'^2}$
Line distribution	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{l'} \hat{\mathbf{R}}' \dfrac{\rho_\ell \, dl'}{R'^2}$
Infinite sheet of charge	$\mathbf{E} = \hat{\mathbf{z}} \dfrac{\rho_s}{2\varepsilon_0}$
Infinite line of charge	$\mathbf{E} = \dfrac{\mathbf{D}}{\varepsilon_0} = \hat{\mathbf{r}} \dfrac{D_r}{\varepsilon_0} = \hat{\mathbf{r}} \dfrac{\rho_\ell}{2\pi\varepsilon_0 r}$
Dipole	$\mathbf{E} = \dfrac{qd}{4\pi\varepsilon_0 R^3} (\hat{\mathbf{R}} 2\cos\theta + \hat{\boldsymbol{\theta}} \sin\theta)$
Relation to *V*	$\mathbf{E} = -\nabla V$

PROBLEMS

Sections 4-2: Charge and Current Distributions

*4.1 A cube 2 m on a side is located in the first octant in a Cartesian coordinate system with one of its corners at the origin. Find the total charge contained in the cube if the charge density is given by $\rho_v = xy^2 e^{-2z}$ (mC/m^3).

4.2 Find the total charge contained in a cylindrical volume defined by $r \le 2$ m and $0 \le z \le 3$ m if $\rho_v = 20rz$ (mC/m^3).

*4.3 Find the total charge contained in a round-top cone defined by $R \le 2$ m and $0 \le \theta \le \pi/4$ given that $\rho_v = 10R^2 \cos^2\theta$ (mC/m^3).

4.4 If the line charge density is given by $\rho_l = 24y^2$ (mC/m), find the total charge distributed on the y axis from $y = -5$ to $y = 5$.

* Answer(s) available in Appendix E.

数学与物理模型

静电场的麦克斯韦方程组

名称	微分形式	积分形式
高斯定律	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
基尔霍夫定律	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

电场

电流密度	$\mathbf{J} = \rho_v \mathbf{u}$
泊松方程	$\nabla^2 V = -\dfrac{\rho_v}{\varepsilon}$
拉普拉斯方程	$\nabla^2 V = 0$
电阻	$R = \dfrac{-\int_l \mathbf{E} \cdot d\mathbf{l}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$
边界条件	见表4-3
电容	$C = \dfrac{\int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}}$
RC 关系	$RC = \dfrac{\varepsilon}{\sigma}$
电能密度	$w_e = \frac{1}{2}\varepsilon E^2$

点电荷	$\mathbf{E} = \hat{\mathbf{R}} \dfrac{q}{4\pi\varepsilon R^2}$
多个点电荷	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \sum_{i=1}^{N} \dfrac{q_i(\mathbf{R} - \mathbf{R}_i)}{\lvert\mathbf{R} - \mathbf{R}_i\rvert^3}$
体电荷分布	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{v'} \hat{\mathbf{R}}' \dfrac{\rho_v \, dv'}{R'^2}$
面电荷分布	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{S'} \hat{\mathbf{R}}' \dfrac{\rho_s \, ds'}{R'^2}$
线电荷分布	$\mathbf{E} = \dfrac{1}{4\pi\varepsilon} \int_{l'} \hat{\mathbf{R}}' \dfrac{\rho_\ell \, dl'}{R'^2}$
无限均匀面电荷	$\mathbf{E} = \hat{\mathbf{z}} \dfrac{\rho_s}{2\varepsilon_0}$
无限均匀线电荷	$\mathbf{E} = \dfrac{\mathbf{D}}{\varepsilon_0} = \hat{\mathbf{r}} \dfrac{D_r}{\varepsilon_0} = \hat{\mathbf{r}} \dfrac{\rho_\ell}{2\pi\varepsilon_0 r}$
电偶极子	$\mathbf{E} = \dfrac{qd}{4\pi\varepsilon_0 R^3} (\hat{\mathbf{R}} 2\cos\theta + \hat{\boldsymbol{\theta}} \sin\theta)$
与电势 V 的关系	$\mathbf{E} = -\nabla V$

说明：面电荷位于 z=0；表中方向对应 z>0，z<0 时电场反向。电偶极子公式适用于 R 远大于电荷间距 d。

问题

第 4-2 节：电荷和电流分布

*4.1 边长为 2 m 的正方体位于笛卡尔坐标系中的第一个八分圆内，其一个角位于原点。如果电荷密度由 $\rho_v = xy^2 e^{-2z}$ (mC/m^3) 给出，则求立方体中包含的总电荷。

4.2 求由 $r \le 2$ m 和 $0 \le z \le 3$ m if $\rho_v = 20rz$ (mC/m^3) 定义的圆柱形体积中包含的总电荷。

*4.3 给定 $\rho_v = 10R^2 \cos^2\theta$ (mC/m^3)，求 $R \le 2$ m 和 $0 \le \theta \le \pi/4$ 定义的圆顶锥体中包含的总电荷。

4.4 如果线电荷密度由 $\rho_l = 24y^2$ (mC/m) 给出，求分布在 y 轴上从 $y = -5$ 到 $y = 5$ 的总电荷。

* 标记的习题在附录 E 提供答案。

Important Terms Provide definitions or explain the meaning of the following terms:

boundary conditions
capacitance C
charge density
conductance G
conduction current
conductivity σ
conductor
conservative field
constitutive parameters
convection current
Coulomb's law
current density $\mathbf{J}$
dielectric breakdown voltage V_{br}
dielectric material
dielectric strength E_{ds}
dipole moment $\mathbf{p}$
electric dipole
electric field intensity $\mathbf{E}$
electric flux density $\mathbf{D}$
electric potential V
electric susceptibility χ_e
electron drift velocity $\mathbf{u}_e$
electron mobility μ_e
electrostatic energy density w_e
electrostatic potential energy W_e
electrostatics
equipotential
Gaussian surface
Gauss's law
hole drift velocity $\mathbf{u}_h$
hole mobility μ_h
homogeneous material
image method
isotropic material
Joule's law
Kirchhoff's voltage law
Laplace's equation
linear material
Ohm's law
perfect conductor
perfect dielectric
permittivity ε
Poisson's equation
polarization vector $\mathbf{P}$
relative permittivity ε_r
semiconductor
static condition
superconductor
volume, surface, and line charge densities

4.5 Find the total charge on a circular disk defined by $r \le a$ and $z = 0$ if:

(a) $\rho_s = \rho_{s0}\cos\phi$ (C/m^2)

(b) $\rho_s = \rho_{s0}\sin^2\phi$ (C/m^2)

(c) $\rho_s = \rho_{s0}e^{-r}$ (C/m^2)

(d) $\rho_s = \rho_{s0}e^{-r}\sin^2\phi$ (C/m^2)

where ρ_{s0} is a constant.

4.6 If $\mathbf{J} = \hat{\mathbf{y}}4xz$ (A/m^2), find the current I flowing through a square with corners at $(0,0,0)$, $(2,0,0)$, $(2,0,2)$, and $(0,0,2)$.

4.7 If $\mathbf{J} = \hat{\mathbf{R}}5/R$ (A/m^2), find I through the surface $R = 5$ m.

4.8 An electron beam shaped like a circular cylinder of radius r_0 carries a charge density given by

$$\rho_v = \left(\frac{-\rho_0}{1+r^2}\right) \quad (\text{C/m}^3)$$

where ρ_0 is a positive constant and the beam's axis is coincident with the z axis.

(a) Determine the total charge contained in length L of the beam.

(b) If the electrons are moving in the $+z$ direction with uniform speed u, determine the magnitude and direction of the current crossing the z-plane.

4.9 A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z axis and the electron charge density is given by

$$\rho_v = -cr^2 \qquad (\text{c/m}^3)$$

where c is a constant and r is the radial distance from the axis of the beam.

****(a)** Determine the charge density per unit length.

(b) Determine the current crossing the z-plane.

4.10 A line of charge of uniform density ρ_ℓ occupies a semicircle of radius b as shown in **Fig. P4.10**. Use the material presented in Example 4-4 to determine the electric field at the origin.

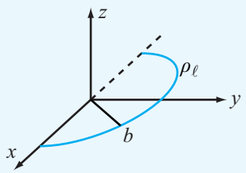


Figure P4.10 Problem 4.10.

Important Terms Provide definitions or explain the meaning of the following terms:

boundary conditions
capacitance C
charge density
conductance G
conduction current
conductivity σ
conductor
conservative field
constitutive parameters
convection current
Coulomb's law
current density $\mathbf{J}$
dielectric breakdown voltage V_{br}
dielectric material
dielectric strength E_{ds}
dipole moment $\mathbf{p}$
electric dipole
electric field intensity $\mathbf{E}$
electric flux density $\mathbf{D}$
electric potential V
electric susceptibility χ_e
electron drift velocity $\mathbf{u}_e$
electron mobility μ_e
electrostatic energy density w_e
electrostatic potential energy W_e
electrostatics
equipotential
Gaussian surface
Gauss's law
hole drift velocity $\mathbf{u}_h$
hole mobility μ_h
homogeneous material
image method
isotropic material
Joule's law
Kirchhoff's voltage law
Laplace's equation
linear material
Ohm's law
perfect conductor
perfect dielectric
permittivity ε
Poisson's equation
polarization vector $\mathbf{P}$
relative permittivity ε_r
semiconductor
static condition
superconductor
volume, surface, and line charge densities

4.5 求由 r ≤ a 和 z = 0 定义的圆盘上的总电荷，如果：

(a) $\rho_s = \rho_{s0}\cos\phi$ (C/m^2) (b) $\rho_s = \rho_{s0}\sin^2\phi$ (C/m^2) (c) $\rho_s = \rho_{s0}e^{-r}$ (C/m^2) (d) $\rho_s = \rho_{s0}e^{-r}\sin^2\phi$ (C/m^2)

其中 ρ_{s0} 是常数。

4.6 如果 J = ŷ4xz (A/m^2)，求流经角点为 (0, 0, 0)、(2, 0, 0)、(2, 0, 2) 和 (0, 0, 2) 的正方形的电流 I。

4.7 如果 J = R̂5/R (A/m^2)，求 I 穿过表面 R = 5 m。

4.8 形状像半径为 r_0 的圆柱体的电子束携带的电荷密度为：

$$\rho_v = \left(\frac{-\rho_0}{1+r^2}\right) \quad (\text{C/m}^3)$$

其中 ρ_0 是正常数，束轴与 z 轴重合。

(a) 确定束流长度 L 中包含的总电荷。 (b) 如果电子以均匀速度 u 沿 +z 方向移动，请确定穿过 z 平面的电流的大小和方向。

4.9 半径为 a 的圆形电荷束由沿 +z 方向以恒定速度 u 移动的电子组成。

电子束的轴与 z 轴重合，电子电荷密度由下式给出

$$\rho_v = -cr^2 \qquad (\text{c/m}^3)$$

其中 c 是常数，r 是距束轴的径向距离。

*(a) 确定每单位长度的电荷密度。 (b) 确定穿过 z 平面的电流。

4.10 均匀密度ρ_ℓ的电荷线占据半径为b的半圆，如图P4.10所示。使用例 4-4 中提供的材料来确定原点处的电场。

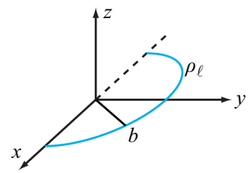


图 P4.10 问题 4.10。

Section 4-3: Coulomb's Law

*4.11 A square with sides of 2 m has a charge of 40 μC at each of its four corners. Determine the electric field at a point 5 m above the center of the square.

4.12 Three point charges, each with $q = 3$ nC, are located at the corners of a triangle in the x–y plane, with one corner at the origin, another at $(2\text{ cm}, 0, 0)$, and the third at $(0, 2\text{ cm}, 0)$. Find the force acting on the charge located at the origin.

*4.13 Charge $q_1 = 6$ μC is located at $(1\text{ cm}, 1\text{ cm}, 0)$ and charge q_2 is located at $(0, 0, 4\text{ cm})$. What should q_2 be so that $\mathbf{E}$ at $(0, 2\text{ cm}, 0)$ has no y component?

4.14 A line of charge with uniform density $\rho_\ell = 8$ (μC/m) exists in air along the z axis between $z = 0$ and $z = 5$ cm. Find $\mathbf{E}$ at (0,10 cm,0).

4.15 Electric charge is distributed along an arc located in the x–y plane and defined by $r = 2$ cm and $0 \le \phi \le \pi/4$. If $\rho_\ell = 5$ (μC/m), find $\mathbf{E}$ at $(0, 0, z)$ and then evaluate it at

*(a) the origin,

(b) $z = 5$ cm, and

(c) $z = -5$ cm.

4.16 A line of charge with uniform density ρ_l extends between $z = -L/2$ and $z = L/2$ along the z axis. Apply Coulomb's law to obtain an expression for the electric field at any point $P(r, \phi, 0)$ on the x–y plane. Show that your result reduces to the expression given by Eq. (4.33) as the length L is extended to infinity.

*4.17 Repeat Example 4-5 for the circular disk of charge of radius a, but in the present case, assume the surface charge density to vary with r as

$$\rho_s = \rho_{s0} r^2 \qquad (\text{C/m}^2)$$

where ρ_{s0} is a constant.

4.18 Multiple charges at different locations are said to be in equilibrium if the force acting on any one of them is identical in magnitude and direction to the force acting on any of the others. Suppose we have two negative charges: one located at the origin and carrying charge $-9e$ and the other located on the positive x axis at a distance d from the first one and carrying charge $-36e$. Determine the location, polarity, and magnitude of a third charge whose placement would bring the entire system into equilibrium.

Section 4-4: Gauss's Law

*4.19 Three infinite lines of charge, all parallel to the z axis, are located at the three corners of the kite-shaped arrangement shown in Fig. P4.19. If the two right triangles are symmetrical and of equal corresponding sides, show that the electric field is zero at the origin.

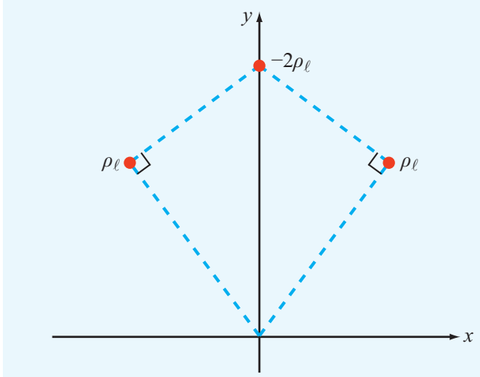


Figure P4.19 Kite-shaped arrangement of line charges for Problem 4.19.

4.20 Three infinite lines of charge, $\rho_{l_1} = 3$ (nC/m), $\rho_{l_2} = -3$ (nC/m), and $\rho_{l_3} = 3$ (nC/m) are all parallel to the z axis. If they pass through the respective points $(0, -b)$, $(0, 0)$, and $(0, b)$ in the x–y plane, find the electric field at $(a, 0, 0)$. Evaluate your result for $a = 2$ cm and $b = 1$ cm.

4.21 A horizontal strip lying in the x–y plane is of width d in the y direction and infinitely long in the x direction. If the strip is in air and has a uniform charge distribution ρ_s, use Coulomb's law to obtain an explicit expression for the electric field at a point P located at a distance h above the centerline of the strip. Extend your result to the special case where d is infinite and compare it with Eq. (4.25).

4.22 Given the electric flux density

$$\mathbf{D} = \hat{\mathbf{x}} 2(x + y) + \hat{\mathbf{y}}(3x - 2y) \qquad (\text{C/m}^2),$$

determine

(a) ρ_v by applying Eq. (4.26),

(b) the total charge Q enclosed in a cube 2 m on a side, located in the first octant with three of its sides coincident with the x, y, and z axes and one of its corners at the origin, and

(c) the total charge Q in the cube, obtained by applying Eq. (4.29).

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第 4-3 节：库仑定律

*4.11 边长为 2 m 的正方形，其四个角的电荷均为 40 μ C 。确定正方形中心上方 5 m 处的电场。

4.12 三个点电荷，每个 q = 3 nC，位于 x-y 平面中三角形的角上，其中一个角位于原点，另一个角位于 (2 cm, 0, 0) ，第三个角位于 (0, 2 cm, 0)。求作用在原点电荷上的力。

*4.13 电荷 q_1 = 6 μ C 位于 (1 cm, 1 cm, 0)，电荷 q_2 位于 (0 , 0, 4 cm)。 q_2 应该是多少，以便 E 在 (0, 2 cm, 0) 处没有 y 分量？

4.14 空气中存在一条密度均匀的电荷线 $\rho_\ell = 8$ (μ C/m)，沿 z 轴位于 z = 0 和 z = 5 cm 之间。在 (0,10 cm,0) 处找到 E。

4.15 电荷沿 x-y 平面内的弧分布，由 r = 2 cm 和 $0 \le \phi \le \pi/4$ 定义。如果 $\rho_\ell = 5$ (μ C/m)，则在 (0, 0, z) 处找到 E，然后在

*(a) 原点，

(b) z = 5 厘米，以及

(c) z = −5 厘米。

4.16 具有均匀密度 ρ_l 的电荷线沿 z 轴在 z = −L/2 和 z = L/2 之间延伸。应用库仑定律获得 x-y 平面上任意点 P(r, ϕ , 0) 处的电场表达式。证明你的结果可以简化为方程式给出的表达式。 (4.33) 当长度 L 扩展到无穷大时。

*4.17 对半径为 a 的电荷圆盘重复例 4-5，但在本例中，假设表面电荷密度随 r 变化为

$$\rho_s = \rho_{s0} r^2 \qquad (\text{C/m}^2)$$

其中 ρ_{s0} 是常数。

4.18 如果作用在其中任何一个电荷上的力与作用在其他任何一个电荷上的力的大小和方向相同，则称不同位置的多个电荷处于平衡状态。假设我们有两个负电荷：一个位于原点并携带电荷 −9e，另一个位于正 x 轴上，距第一个负电荷距离 d，并携带电荷 −36e。确定第三个电荷的位置、极性和大小，该电荷的放置将使整个系统达到平衡。

第 4-4 节：高斯定律

*4.19 三个无限电荷线均平行于 z 轴，位于风筝形排列的三个角处

如图P4.19所示。如果两个直角三角形对称且对应边相等，则表明原点处电场为零。

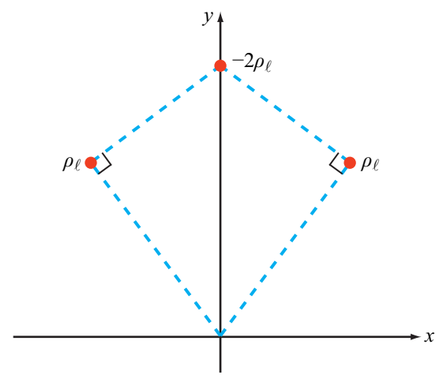


图 P4.19 问题 4.19 中线电荷的风筝形排列。

4.20 三个无限电荷线 $\rho_{l_1} = 3$ (nC/m)、$\rho_{l_2} = -3$ (nC/m) 和 $\rho_{l_3} = 3$ (nC/m) 均平行于 z 轴。如果它们穿过 x-y 平面上的相应点 (0, −b)、(0, 0) 和 (0, b)，则求 (a, 0, 0) 处的电场。评估 a = 2 厘米和 b = 1 厘米的结果。

4.21 位于 x-y 平面上的水平条带在 y 方向上的宽度为 d，在 x 方向上的长度无限长。如果带材在空气中并且具有均匀的电荷分布 ρ_s，则使用库仑定律获得位于带材中心线上方距离 h 处的点 P 处的电场的显式表达式。将你的结果扩展到 d 无穷大的特殊情况，并将其与方程进行比较。 （4.25）。

4.22 给定电通量密度

$$\mathbf{D} = \hat{\mathbf{x}} 2(x + y) + \hat{\mathbf{y}}(3x - 2y) \qquad (\text{C/m}^2),$$

决定

(a) ρ_v 通过应用等式。 (4.26)，(b) 包含在边长 2 m 的立方体中的总电荷 Q，该立方体位于第一个八分圆内，其三个边与 x、y 和 z 轴重合，且其中一个角位于原点，以及 (c) 通过应用式 (4.26) 获得的立方体中的总电荷 Q。 （4.29）。